

No. of Functional Features Included in the Solution

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Weather Data Input	Accepts meteorological and rainfall values from users	Predict Page	Done	High
2	Data Validation	Validates all user inputs before prediction	Flask Backend + JavaScript	Done	High
3	Flood Prediction	Predicts flood occurrence using Random Forest model	Machine Learning Model	Done	High
4	Confidence Score	Displays prediction confidence percentage	Prediction Engine	Done	Medium
5	Feature Contribution Analysis	Explains which parameters influenced prediction	Explanation Module	Done	High
6	Interactive Dashboard	Displays results with gauge and Chart.js visualization	Result Page	Done	Medium
7	Responsive User Interface	Modern Glassmorphism interface for all devices	HTML/CSS/Bootstrap	Done	Medium
8	IBM Cloud Deployment Support	Deployment-ready configuration using Procfile and runtime	Deployment Module	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Flood Prediction, Data Validation, ML Model, Confidence Score, Feature Explanation

Additional / Nice-to-Have Features	☐ Interactive confidence gauge visualization ☐ Feature contribution charts using Chart.js ☐ Responsive Glassmorphism user interface ☐ IBM Cloud deployment support ☐ Modular architecture for easy maintenance
Features Tested & Verified	☐ User input validation ☐ Machine Learning prediction accuracy ☐ Flood/No Flood classification ☐ Confidence score generation ☐ Feature explanation module ☐ Flask routing and page navigation ☐ Responsive UI on desktop and mobile

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Dashboard
2	Backend / Logic	4	Flask Routing, Input Validation, Prediction Engine, Explanation Generator
3	Database / Storage	2	Saved ML Model (best_model.pkl), Scaler (scaler.pkl)
4	API / Integration	1	Flask API for Prediction Requests
5	Security / Authentication	1	Input Validation and Error Handling